



1 Pipeline Breakdown

- Ingestion

Accepts both **structured** (CSV/JSON ATS exports) and **unstructured** (PDF/DOCX/TXT resumes, LinkedIn/GitHub URL text) sources. Each source is tagged with a `source_type` and `source_id` before entering the pipeline so provenance can be tracked end-to-end.
- Extraction

Deterministic regex/heuristic entity extraction (no LLM) for name, email, phone, location, skills, dates. Pre-cleans OCR artifacts (zero-width spaces, broken `mailto:` tags) and strips URLs before phone parsing to avoid false-positive numbers from LinkedIn/GitHub IDs.
- Normalization

Converts every field into the canonical format below regardless of source quirks — this is what makes merging safe.
- Merge / Confidence

Groups records belonging to the same candidate, applies the conflict-resolution policy (Section 3), and computes a per-field confidence score.
- Projection

Reshapes the internal canonical record into the default schema or any custom runtime `output_config` (rename/flatten/omit) — the master record is never mutated.
- Validation

Schema-checks the projected JSON before returning it; degrades gracefully (nulls + low confidence) rather than failing the whole record on a bad source.

2 Canonical Output Schema & Normalized Formats

```
{
  "candidate_id": "sha1(name+email)",
  "full_name": "string",
  "emails": ["lowercase"],
  "phones": ["E.164"],
  "location": { "country": "ISO-3166 alpha-2" },
  "skills": ["canonical, deduped"],
  "experience": [{ "start": "YYYY-MM", "end": "YYYY-MM|null" }],
  "provenance": {
    "<field>": { "source", "confidence", "timestamp" }
  }
}
```

- Dates** → `YYYY-MM`; partial dates padded, "Present" → `null` end.
- Phones** → parsed via `phonenumbers` lib, validated + formatted `E.164`.
- Emails** → lowercased, trimmed, zero-width-space stripped.
- Skills** → alias map (`ml-Machine Learning`) + case-fold dedupe.
- Location** → free text → ISO-3166 alpha-2 via lookup + fuzzy fallback.

3 Merge & Conflict-Resolution Policy

Step	Rule
Match key	Exact-word intersection on normalized full name + at least one shared email/phone — prevents "John S" merging into "John Smith".
Source priority	Static configurable rank, default <code>Resume > LinkedIn > ATS CSV > GitHub</code> . Highest-priority non-null value wins scalar fields.
Arrays	Skills / emails / phones are unioned + deduplicated across all sources, not overwritten.
Confidence	<code>1.0</code> if ≥2 sources agree, <code>0.7</code> single high-priority source, <code>0.4</code> low-priority-only or fuzzy-matched.
Provenance	Every field stores <code>{source, confidence, timestamp}</code> — full audit trail, nothing is silently overwritten.

4 Custom Output Config

Runtime JSON/YAML config drives the Projection layer: `fields[]` (path + rename), `include_provenance`, and `on_missing` (`omit | null | default`). Validated against a JSON-Schema for the target shape before being returned — the internal canonical record is read-only input to this step, so re-projecting never loses data.

5 Edge Cases Handled (and what's deliberately descoped)

- Cross-candidate bleeding

— exact-word match key blocks "John S" → "John Smith" false merges.
- Invisible OCR glitches

— strips `\u200b` & malformed `mailto:` before regex.
- URL false positives

— strips URLs pre-phone-parse so LinkedIn/GitHub IDs aren't read as numbers.
- Filename heuristics

— extracts names from filenames while blacklisting titles (AI, Engineer, Developer).
- Malformed CSV rows

— skipped individually with a warning; batch ingestion never aborts.
- Missing/garbage source

— field degrades to `null` + low confidence rather than failing the record.
- Deliberately out of scope (time-boxed):

fuzzy cross-source name matching beyond exact-word intersection, multi-language resume parsing, real-time streaming ingestion, and PII encryption-at-rest — all flagged in the README as v2 follow-ups rather than silently dropped.